Assignment: Image Denoising

Draw, implement and evaluate deep-learning-based image denoising models and compare two different learning strategies: direct learning and residual learning.

Models to draw and implement:

1. Denoising Autoencoder (DAE)
2. RED-Net
3. U-Net
4. DnCNN
5. ***Your proposed model for direct denoising (say Model1)***
6. ***Your proposed model for residual denoising (say Model2)***

Dataset:

Use the dataset prepared by all of you.

Use at least one type of noise, such as Gaussian noise, salt-and-pepper noise, or speckle noise. You may also experiment with different noise levels.

Divide the dataset into training, validation, and test sets. Use the same dataset and data split for all four models.

Models:

Draw the architecture and implement the following models that receive a noisy image as input and directly reconstruct the corresponding clean image.

Input: Noisy image y
Target: Clean image x
Output: Denoised image x_hat

1. Denoising Autoencoder (DAE)
2. RED-Net
3. U-Net
4. DnCNN
5. Model1
6. Model2

Training:

Train all four models using the same training dataset and, as far as possible, the same training conditions.

For direct-learning models, use the clean image as the target.
For residual-learning models, use the residual noise as the target.
You may use MSE, L1 loss, or another suitable image-restoration loss.

Evaluation:

Evaluate all models on the same test set.

Use the following metrics:

1. MSE (Mean Squared Error)
2. PSNR (Peak Signal-to-Noise Ratio)
3. SSIM (Structural Similarity Index Measure)

Compare the denoised images with the corresponding clean images.

Required Experiments:

Experiment 1: Model Comparison

Compare the denoising performance of:
DAE vs. RED-Net vs. U-Net vs. DnCNN vs. Model1 vs. Model2

Experiment 2: Direct vs. Residual Learning

Compare the performance of the direct-learning models with the residual-learning model.
Discuss whether residual learning provides any advantage over direct learning.

Experiment 3: Different Noise Levels

Test the models using at least three different noise levels.

For example:

$\sigma = 15$
$\sigma = 25$
$\sigma = 50$

Analyze how the performance of each model changes as the noise level increases.

Experiment 4: Visual Comparison

For several test images, show:

- Clean image
- Noisy image
- DAE output

- RED-Net output
- U-Net output
- DnCNN output
- Model1 output
- Model2 output

Discuss noise removal, edge preservation, texture preservation, blurring, and artifacts.

Results:

Prepare a table containing:

Model | Learning Type | Target | MSE | PSNR | SSIM

Also compare the computational requirements of the models, including training time, inference time, and number of parameters if possible.

Discussion:

Answer the following questions:

1. What do you understand by ‘residual’ and ‘residual in image denoising’?
2. Why does DnCNN predict the residual noise instead of directly predicting the clean image?
3. What are the advantages and disadvantages of direct learning?
4. What are the advantages and disadvantages of residual learning?
5. Is there any difference between ‘skip connection’ and ‘residual’?
6. How do the skip connections of RED-Net and U-Net differ?
7. Why does U-Net use concatenation in its skip connections?
8. Why does RED-Net use additive skip connections?
9. Which model achieves the best MSE, PSNR, SSIM?
10. Which model achieves the best PSNR?
11. Which model achieves the best SSIM?
12. Which model provides the best visual denoising quality?
13. Which model preserves edges and textures most effectively?
14. Does residual learning outperform direct learning in your experiments?
15. Does the relative performance change when the noise level increases?

Conclusion:

Conclude the assignment by discussing which model performed best and whether direct learning or residual learning was more effective for the tested denoising problem.

Important:

RED-Net and U-Net are network architectures, whereas direct learning and residual learning describe the learning target. Therefore, the comparison should distinguish between architectural differences and learning-target differences. Do not attribute all performance differences between the models solely to direct versus residual learning.